Sketch-Based Region Adaptive Sparse Unmixing Applied to Hyperspectral Image

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***Abstract*— Hyperspectral image (HSI) unmixing is an important issue of research due to its effect on the subsequent processing of HSIs. Recently, the sparse regression method with spatial information has been successfully applied in hyperspectral unmixing (HU). However, most sparse regression methods ignore the difference in spatial structure handling with only one sparse constraint. In fact, the pixels in detail regions are more likely to be severely mixed with more endmembers participated, and the sparsity degree of its corresponding abundances is relatively low. Considering the sparsity difference of abundances, a sketch-based region adaptive sparse unmixing applied to HSI is proposed in this article. Inspired by the vision computing theory, we use the region generation algorithm based on a sketch map to differentiate the homogeneous regions and detail regions. Then, the abundances of these two kind regions in HSIs are separately constrained by sparse regularizers of $L_{1/2}$ and L_1 with a proposed manifold constraint. Our method not only makes full use of the spatial information in HSIs but also exploits the latent structure of data. The encouraging experimental results on three data sets validate the effectiveness of our method for HU.**

***Index Terms*— Hyperspectral sparse unmixing, manifold constraint, nonnegative matrix factorization (NMF), region difference, sketch map.**

I. Introduction

With the development of imaging spectroscopy, hyperspectral images (HSIs) acquired by high-spectral-resolution imaging sensors have received more and more attention [1]–[3]. Due to the rich spectral information containing hundreds of contiguous and narrow spectral bands, hyperspectral imaging has been widely applied in many fields, such as agriculture, pollution detection, and food quality inspection [4]–[6].

Due to the limited spatial resolution of sensors and the variant ground surface, HSIs always exist the mixed pixels that are the mixtures of multiple ground objects [7]–[9]. The problem of mixed pixels will decrease the performance of processing methods for HSIs. Therefore, hyperspectral unmixing (HU) has become an essential step for HSIs' analysis. The task of HU aims to decompose a mixed pixel into a collection of constituent spectra (called endmember) and their relative proportions (also called abundance).

There are two kinds of spectral mixture models for HU, i.e., linear spectral mixture models (LMMs) and non-LMMs. LMMs are widely adopted, which are based on the assumption that the mixing scale is macroscopic and distinct endmembers are not interfered with each other. Since it is easy to be solved and can satisfy the basic requirement of research, LMMs have been one popular model for HU. In addition, two important constraints for HU are the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). Under the LMMs, nonnegative constrained least squares and fully constrained least squares (FCLS) [10] are proposed with different abundance constraints. Generally, linear HU methods can roughly be classified into three categories: geometrical methods, statistical methods, and sparse regression methods [9].

Convention algorithms for HU have two important steps: endmember extraction and abundance estimation. For endmember extraction, the first step is to estimate the number for endmember, which is a key process in HU. The commonly used methods of endmember number estimation are hyperspectral signal identification by minimum error (Hysime) [11], minimum noise fraction [12], virtual dimensionality (VD) [13], principal component analysis [14], and so on. In addition, the algorithms for identifying endmembers have been proposed in recent years, including N-FINDR [15], vertex component analysis (VCA) [16], pixel purity index [17], alternating volume maximization [18], minimum volume simplex analysis [19], and simplex projection method [20]. These mentioned methods for endmember extraction are some typical geometrical methods, which assume that pixels of HSIs are located within a simplex, whose vertices are corresponding to the endmembers. Unfortunately, these geometrical methods require the presence of pure pixels in the original image, which will usually be invalid in practice. When the spectral mixtures are highly mixed, the statistical methods of endmember extraction are a powerful alternative with higher computational complexity compared with the geometrical methods. Some representative methods based on statistic are in [21]–[24].

Nonnegative matrix factorization (NMF) [25], which decomposes the mixed pixels into endmembers and

Manuscript received July 22, 2019; revised December 28, 2019 and March 23, 2020; accepted April 16, 2020. Date of publication May 14, 2020; date of current version November 24, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61772400, Grant 61801351, Grant 61772399, Grant 61902298, and Grant 91438201, in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-08, in part by the China Postdoctoral Science Foundation Funded Project under Grant 2017M613081, and in part by the Fundamental Research of the Central University under Grant JB191911. *(Corresponding author: Xiangrong Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2020.2991194

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abundances simultaneously, can capture the underlying structure of data in low-dimensional representation space. However, NMF is an ill-posed problem whose cost function is not convex, making the solution space very large. To improve the performance of NMF and reduce the solution space, researchers impose some prior knowledge as further constraints, such as semi-NMF, nonsmooth NMF [26], multilayer NMF [27], and sparse NMF [28], [29]. Qian *et al.* [30] utilized the $L_{1/2}$-regularizer constraint on abundances for NMF ($L_{1/2}$-NMF), which leads to a more satisfactory unmixing result. He *et al.* [31] proposed a weighted sparse regularizer for the NMF algorithm whose weight is related to the abundance and updated adaptively. Besides, robust collaborative NMF (CoNMF) for HU [32] adds some spatial constraints for abundances, which performs three steps of the HU chain, i.e., the estimation of the number of endmembers, endmembers' identification, and fractional abundance estimation. Graph regularized $L_{1/2}$-NMF (GLNMF) aims at seizing the internal manifold structure of hyperspectral data with sparse constraint and graph regularization [33]. Due to its good mathematical foundation, a graph-based method has been widely studied even for some other tasks [34], [35]. Most of these mentioned methods consider the sparse constraint in the NMF model, and unmixing methods based on sparse regression also take the sparse characteristic of hyperspectral data into account.

Recently, due to the wide availability of spectral libraries, sparse unmixing has been shown to be able to circumvent the drawbacks introduced by such virtual endmembers and the unavailability of pure pixels [36], [37]. The sparse unmixing algorithm via variable splitting and augmented Lagrangian (SUnSAL) [38] adopts the L_1 regularizer on the abundance matrix, which assumes that the number of endmembers participated in each pixel is low. Besides, the collaborative SUnSAL [39] introduces the $L_{2,1}$ regularizer to constrain the pixels in HSIs to share the same active set of endmembers. The sparse unmixing via variable splitting augmented Lagrangian and total variation [40] takes the spatial information into consideration, which promotes piecewise transitions of abundance estimation in sparse unmixing. By nonlocal Euclidean medians filtering approach, Feng *et al.* [41] proposed an adaptive nonlocal Euclidean medians sparse unmixing method that utilized a joint maximum *a posteriori* strategy.

However, most of the abovementioned methods are operated at the pixel level and ignore the spatial information in HSIs which is an important feature for HU. Tong *et al.* [42] introduced a novel region-based structure-preserving NMF to explore consistent data distribution in the same region while discriminating different data structures across regions in the unmixing data. Due to the region-based operation, the results show better performance than the methods implemented on pixels. Zhang *et al.* [43] combined two different spectral mixture models for HU by considering the different characteristics of regions. These region-based methods also have the characteristics of reducing the influence of noises by adopting the spatial neighbor information in HSIs. Shi and Wang [44] summarized the available spatial–spectral unmixing methods and demonstrated the advantages of unmixing method integrating both spatial information and spectral information. Xu *et al.* [45] presented a new spatial preprocessing method derived from the simple linear iterative clustering method while incorporating the spatial correlation and the spectral similarity for endmember extraction. Uezato *et al.* [46] developed an HSI unmixing method with spatial regularization which integrated the height information of LiDAR data. In recent years, a number of studies have revealed that using spatial information by considering the existing spatial autocorrelation among the pixels can improve the performance for the unmixing [44].

Inspired by the advantages of spatial sparse information in HSIs, we develop a region-based sparse unmixing framework for HSIs. The assumption of this article is that most pixels in the image are mixed [47], [48], and the pixels in the detail regions are more likely to be heavily mixed with more endmembers participated than in the homogeneous regions. Therefore, it is reasonably inferred that the sparsity degree of the abundances for these two regions is obviously different, and the sparsity degree of abundances in homogeneous regions is relatively high. However, most sparse unmixing methods ignore the sparsity difference with only one sparse constraint. Thus, sketch-based region adaptive sparse unmixing (SRASU) applied to HSI is proposed in this article. First, inspired by the vision computing theory, we use the region generation algorithm based on the sketch map to distinguish the homogeneous regions and detail regions of HSIs. Then, the abundances of these two regions are separately constrained by different sparse regularizers. Besides, to exploit the latent structure of hyperspectral data, manifold regularizer constraint constructed by a novel function is also adopted. The encouraged results of experiments on three data sets illustrate the effectiveness of SRASU. The whole process of our method is shown in Fig. 1. The main contribution of this article is as follows.

1) The proposed method takes the sparsity degree difference into account and separately constrains the abundances of homogeneous regions and detail regions by different sparse regularizers.
2) A novel algorithm named manifold sparse constraint based on NMF (MSNMF) for HU is proposed in this article by considering the intrinsic manifold structure of data.
3) It is the first time to use the region generation algorithm based on the sketch map for HSIs unmixing in which the vision computing information is made full use of.

The rest of this article is organized as follows. In Section II, we describe the LMMs and NMF algorithm briefly. Section III gives a detailed introduction of the proposed method SRASU. The experimental results as well as comparisons and analyses are shown in Section IV, followed by conclusions in Section V.

II. Related Work

A. LMMs

LMMs are one of the most popular models that have been widely used in HU. Let $\mathbf{y}_i \in \Re^{L\times 1}$ denote the ith pixel of hyperspectral data matrix with L spectral bands, $\mathbf{E} \in \Re^{L\times K}$ indicate the endmember matrix with K endmembers,

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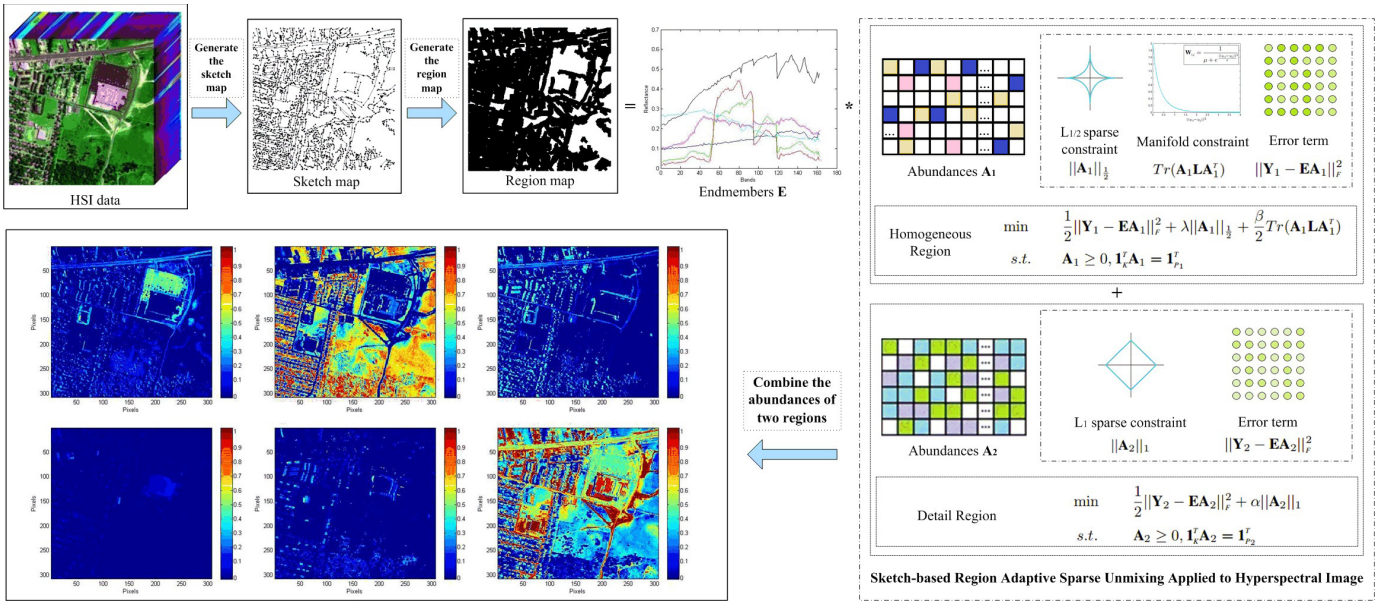


Fig. 1. Illustration of the proposed method SRASU.

$\mathbf{a}_i = [a_{1i}, \ldots, a_{Ki}]^T \in \Re^{K\times 1}$ stand for the abundance fractions of the ith pixel, and $\mathbf{n}_i \in \Re^{L\times 1}$ represent the additional noise. LMMs can be written as

$$\mathbf{y}_i = \mathbf{E}\mathbf{a}_i + \mathbf{n}_i \,. \tag{1}$$

The aforementioned two constraints for abundance, i.e., ANC and ASC, separately are $a_{ki} \geq 0$, $\sum_{k=1}^{K} a_{ki} = 1$. If $\mathbf{Y} \in \Re^{L\times P}$ is treated as the total observed image with P pixels, $\mathbf{A} \in \Re^{K\times P}$ composed of $\mathbf{a}_i$ is the abundance matrix, and $\mathbf{N} \in \Re^{L\times P}$ represents noise matrix, then the matrix form of LMMs is achieved: $\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N}$.

From the LMMs, we can find that the task of unmixing is to factorize a high-dimensional matrix $\mathbf{Y}$ into two low-dimensional matrices $\mathbf{E}$ and $\mathbf{A}$, subjected to ANC and ASC.

B. NMF

NMF built in terms of LMMs is fairly attractive due to its great success in many applications. The objective function of NMF based on the Euclidean distance is as follows:

$$\min_{E,A} \frac{1}{2}||\mathbf{Y} - \mathbf{E}\mathbf{A}||_F^2 \quad \text{s.t. } \mathbf{E} \geq 0, \quad \mathbf{A} \geq 0 \tag{2}$$

where the operator $||\cdot||_F$ denotes the Frobenius norm. To minimize the objective function in (2), [25] proposed an iterative algorithm as follows:

$$\mathbf{E} \leftarrow \mathbf{E}. * \mathbf{Y}\mathbf{A}^T ./ \left(\mathbf{E}\mathbf{A}\mathbf{A}^T\right) \tag{3}$$

$$\mathbf{A} \leftarrow \mathbf{A}. * \mathbf{E}^T\mathbf{Y} ./ \left(\mathbf{E}^T\mathbf{E}\mathbf{A}\right) \tag{4}$$

where $(\cdot)^T$ is the transpose of matrix, .* denotes the element-wise multiplication, and ./ represents the elementwise division.

Unfortunately, the objective function of NMF is nonconvex, which leads to a large quantity of local minima. Hence, some constraints have been introduced into standard NMF [28], [49] to reduce feasible solution set. Taking the sparsity of abundance into consideration, the $L_{1/2}$-NMF algorithm has been proposed in [30] whose objective function is

$$\min_{E,A} \frac{1}{2}||\mathbf{Y} - \mathbf{E}\mathbf{A}||_F^2 + \lambda||\mathbf{A}||_{\frac{1}{2}} \quad \text{s.t. } \mathbf{A} \geq 0, \quad \mathbf{1}_K^T\mathbf{A} = \mathbf{1}_P^T \tag{5}$$

where λ is a balance parameter to adjust the sparsity of abundance, $||\mathbf{A}||_{1/2} = \sum(a_{ij})^{1/2}$, and a_{ij} is the abundance fraction of the ith endmember at the jth pixel in HSIs. The rescaled gradient descent in [28] can be achieved by a modification of multiplicative update rules as follows:

$$\mathbf{E} \leftarrow \mathbf{E}. * \mathbf{Y}\mathbf{A}^T ./ \left(\mathbf{E}\mathbf{A}\mathbf{A}^T\right) \tag{6}$$

$$\mathbf{A} \leftarrow \mathbf{A}. * \mathbf{E}^T\mathbf{Y} ./ \left(\mathbf{E}^T\mathbf{E}\mathbf{A} + \frac{\lambda}{2}\mathbf{A}^{-\frac{1}{2}}\right). \tag{7}$$

The $L_{1/2}$ regularizer is closely linked to the minimum volume method through its geometric interpretation [30]. The $L_{1/2}$-NMF algorithm provides a sparse and accurate unmixing result, while it is not robust enough to noise.

III. SRASU Applied to HSI

Due to the correlation of hyperspectral data, most pixels mix with only a few endmembers, which leads to the sparsity of abundance. The sparse model is a powerful technique for processing signals, and its goal is to implement signal representation with a minimum amount of data. Regularization methods are usually utilized to define the sparsity constraint on the abundance. L_0 regularizer accounts for the number of zero elements in the abundance matrix whose solution is an NP-hard optimization problem. Thus, the L_1 regularizer is used to favor sparse solutions and constrain the obtained abundances in the image, including detail regions and homogeneous regions. However, due to its special spatial location, the detail regions are more likely to involve the mixtures with more endmembers participated than the homogeneous regions.

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Accordingly, the abundances of homogeneous regions are more sparser than those of detail regions. Therefore, we adopt the sparse regularizers of $L_{1/2}$ proposed by Xu *et al.* [50], which can further enforce the sparsity to constrain the abundance of homogeneous regions.

In this article, we propose a novel SRASU applied to HSI. By considering the different sparsity degrees for abundances, it is expected that SRASU improves the performance of unmixing problems.

A. Region Generation Based on Sketch Map

Human vision is essentially a process of information processing. Accordingly, Marr [51] proposed a theoretical framework for visual computing, in which the extraction of the initial sketch map is a key step to analyze and understand the scene information contained in the image. Then, Guo *et al.* [52] presented the mathematical theoretical model and the extraction algorithm of sketch map. Inspired by vision computation and its good performance, the sketch map extraction algorithm proposed by Guo *et al.* [52] is applied to the HSI analysis in this article. Then, the rectangular window with direction is designed for the obtained sketch map that is used to distinguish the homogeneous regions $\mathbf{Y}_1 \in \Re^{L\times P_1}$ and detail regions $\mathbf{Y}_2 \in \Re^{L\times P_2}$.

The whole algorithm adopted to obtain the two kinds of regions consists of two parts, namely sketch map extraction and region generation. For the sketch map extraction task, the steps are as follows: 1) construct the basic dictionary of the image; 2) generate brightness change map of the image; 3) obtain initial sketch map by nonmaximum suppression operation; 4) get the sketch map by greedy tracking operation; and 5) extract the final sketch map using a set of graph operators. Considering the characteristics of the brightness change, the basic dictionary composed of a set of filters is constructed by analyzing the geometry, topology, and other information in the image. The sketch map composed of sketch lines with directional characteristics can reflect the structure information of the original image. In addition, the areas that the sketch lines located indicate the areas with great spectral variance. These areas contain a lot of details, such as edges and textures. More details can be found in these articles [52], [53].

For the task of region generation, we use the rectangular window with the direction to divide the image into two different regions. The specific operations are as follows. The center of the rectangular window is aligned with one end of each sketch line. The long side of the rectangular window is parallel to the sketch line. The rectangular window slides along the direction of the sketch line until the center is aligned with the other end. The area that window slides contains detailed information corresponding to the detail regions. The remaining area of an image is considered as the homogeneous regions.

B. Hyperspectral Sparse Unmixing for Homogeneous Regions

In homogeneous regions $\mathbf{Y}_1$, due to the fewer mixtures, the corresponding abundance is relatively sparser compared with other regions, which means that higher sparsity degree is preferred in these regions. Therefore, we use the $L_{1/2}$ regularizer to constrain the abundance $\mathbf{A}_1 \in \Re^{K\times P_1}$ of homogeneous regions to enhance its sparsity. Due to the $L_{1/2}$ regularizer, the $L_{1/2}$-NMF method can enforce the sparsity of abundances. Inspired by $L_{1/2}$-NMF, the sparse constraint with the $L_{1/2}$ regularizer is also adopted in our model. Moreover, in order to exploit the latent manifold structure of hyperspectral data in the feature space, we propose a sparsity-constrained NMF with manifold regularization called MSNMF for HU.

For the homogeneous regions $\mathbf{Y}_1 \in \Re^{L\times P_1}$ with P_1 pixels, we construct a weighted graph on P_1 pixels where each pixel is considered as a node. The main idea is that when the pixels are closer, their corresponding abundances are more similar. Thus, the weight between notes $\mathbf{y}_i$ and $\mathbf{y}_j$ is defined as follows:

$$\mathbf{W}_{ij} = \frac{1}{\mu + e^{\frac{||\mathbf{y}_i - \mathbf{y}_j||^2}{\sigma}}} \tag{8}$$

where $\mu \in (-1, +\infty)$ is the parameter for controlling the maximum and σ is the scaling parameter for $\mathbf{W}_{ij}$. The two parameters are analyzed in Section IV. Then, the abundances constraint is constructed as follows:

$$\begin{aligned}\frac{1}{2}\sum ||\mathbf{a}_i - \mathbf{a}_j||^2 \mathbf{W}_{ij} &= \sum \mathbf{a}_i^T \mathbf{a}_i \mathbf{D}_{ii} - \sum \mathbf{a}_i^T \mathbf{a}_j \mathbf{W}_{ij} \\ &= \mathrm{Tr}\left(\mathbf{A}_1 \mathbf{D} \mathbf{A}_1^T\right) - \mathrm{Tr}\left(\mathbf{A}_1 \mathbf{W} \mathbf{A}_1^T\right) \\ &= \mathrm{Tr}\left(\mathbf{A}_1 \mathbf{L} \mathbf{A}_1^T\right)\end{aligned} \tag{9}$$

where $\mathrm{Tr}(\cdot)$ is the trace of a matrix, $\mathbf{D}$ is a diagonal matrix, $\mathbf{D}_{ii} = \sum \mathbf{W}_{ij}$, and $\mathbf{L} = \mathbf{D} - \mathbf{W}$. From (9), we can easily find that when two pixels are close, the $\mathbf{W}_{ij}$ value is high, that is to say, the corresponding penalty of $||\mathbf{a}_i - \mathbf{a}_j||^2$ is strict, which will make their abundances more closer.

Finally, by the constraints of manifold and sparsity with $L_{1/2}$ regularizer, the whole objective function, consisting of the error reconstruction term, is as follows:

$$\begin{aligned}&\min \ \frac{1}{2}||\mathbf{Y}_1 - \mathbf{E}\mathbf{A}_1||_F^2 + \lambda ||\mathbf{A}_1||_{\frac{1}{2}} + \frac{\beta}{2}\mathrm{Tr}\left(\mathbf{A}_1 \mathbf{L} \mathbf{A}_1^T\right) \\ &\text{s.t. } \mathbf{A}_1 \geq 0, \quad \mathbf{1}_K^T \mathbf{A}_1 = \mathbf{1}_{P_1}^T\end{aligned} \tag{10}$$

where β is the regularization parameter.

Similar to the NMF problem, an iterative algorithm introduced in [33] is used to solve the objective function recorded in (10), and the updating rules are as follows:

$$\mathbf{E} \leftarrow \mathbf{E}. * \mathbf{Y}_1 \mathbf{A}_1^T ./ \left(\mathbf{E}\mathbf{A}_1\mathbf{A}_1^T\right) \tag{11}$$

$$\mathbf{A}_1 \leftarrow \mathbf{A}_1 . * \left(\mathbf{E}^T \mathbf{Y}_1 + \beta \mathbf{A}_1 \mathbf{W}\right) ./ \left(\mathbf{E}^T \mathbf{E} \mathbf{A}_1 + \frac{\lambda}{2}\mathbf{A}_1^{-\frac{1}{2}} + \beta \mathbf{A}_1 \mathbf{D}\right). \tag{12}$$

Moreover, ASC can be achieved by adding a row to the hyperspectral data matrix $\mathbf{Y}_1$ and the endmember matrix $\mathbf{E}$, which are defined as

$$\mathbf{Y}_{1f} = \begin{bmatrix} \mathbf{Y}_1 \\ \varepsilon \mathbf{1}_{P_1}^T \end{bmatrix}, \quad \mathbf{E}_f = \begin{bmatrix} \mathbf{E} \\ \varepsilon \mathbf{1}_K^T \end{bmatrix}$$

and ε is the parameter to control the influence of ASC over the abundance matrix. Considering the ASC, the abundance matrix $\mathbf{A}_1$ is updated through replacing $\mathbf{Y}_1$

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and $\mathbf{E}$ by $\mathbf{Y}_{lf}$ and $\mathbf{E}_f$, whose iterative criterion is as follows:

$$\mathbf{A}_1 \leftarrow \mathbf{A}_1 . * \left(\mathbf{E}_f^T \mathbf{Y}_{lf} + \beta \mathbf{A}_1 \mathbf{W}\right) ./ \left(\mathbf{E}_f^T \mathbf{E}_f \mathbf{A}_1 + \frac{\lambda}{2} \mathbf{A}_1^{-\frac{1}{2}} + \beta \mathbf{A}_1 \mathbf{D}\right). \tag{13}$$

C. Hyperspectral Sparse Unmixing for Detail Regions

In detail regions $\mathbf{Y}_2$, most pixels are generated by various endmembers. Therefore, the sparsity degree of abundance matrix is lower. By adopting the L_1 regularizer to constrain the abundance $\mathbf{A}_2 \in \Re^{K \times P_2}$ of detail regions, we can get the final objective function as follows:

$$\begin{aligned} &\min \ \frac{1}{2} ||\mathbf{Y}_2 - \mathbf{E}\mathbf{A}_2||_F^2 + \alpha ||\mathbf{A}_2||_1 \\ &\text{s.t. } \mathbf{A}_2 \geq 0, \mathbf{1}_K^T \mathbf{A}_2 = \mathbf{1}_{P_2}^T \end{aligned} \tag{14}$$

where the last term is the sparse constraint on the abundance in detail regions and α is a weight parameter. Our algorithm fully considers the spatial difference of data through different abundance constraints on two regions.

In this article, it is assumed that most pixels in the HSI are mixed. The pure signatures are more likely to be present in spatially homogeneous regions [47], [48]. Therefore, the initial endmembers are extracted by the VCA algorithm from the whole image, which includes all possible endmembers. Then, the endmembers are optimized in homogeneous regions for more finely extracting. In order to unify the endmembers of two regions, we use this obtained endmembers for detail regions unmixing. In addition, the function (14) is the same as the objective function of the SUnSAL algorithm. Therefore, the update criterion for $\mathbf{A}_2$ is achieved according to the SUnSAL algorithm [38]. Finally, the abundance matrix $\mathbf{A}$ is obtained by combining $\mathbf{A}_1$ and $\mathbf{A}_2$.

In our SRASU method, we first extract the endmembers from the whole image as the initial endmembers set. Then, the region generation method based on the sketch map is used to find the two kinds of regions with different characteristics. Due to less mixtures in homogeneous regions, the corresponding abundance matrix is sparser than the one of the detail regions. Since the sparse regularizers of $L_{1/2}$ can further enforce the sparsity, it is adopted to constrain the abundance of homogeneous regions. Then, we get its abundance and the updated endmembers. For the detail regions, the classical sparse model with the L_1 regularizer and the updated endmembers are employed. Finally, we obtain the ultimate abundances by merging these two abundance matrices. The proposed algorithm of SRASU is summarized in Algorithm 1.

Algorithm 1 SRASU Applied to HSI

Input: The observed mixture data $\mathbf{Y}$, the endmember number K, the parameters μ, σ and β.
Output: Endmember signature matrix $\mathbf{E}$ and abundance matrix $\mathbf{A}$.
Initialization: $\mathbf{E}$ by VCA algorithm, $\mathbf{A}$ by FCLS algorithm, divide $\mathbf{Y}$ into homogeneous regions $\mathbf{Y}_1$ and detail regions $\mathbf{Y}_2$ by region generation algorithm, divide the $\mathbf{A}$ into $\mathbf{A}_1$ and $\mathbf{A}_2$ accordingly, calculate the manifold matrix $\mathbf{W}$ and $\mathbf{D}$.
Repeat:
Update $\mathbf{E}$ by Eq. (11);
Augment the $\mathbf{Y}_1$, $\mathbf{E}$;
Update $\mathbf{A}_1$ by Eq. (13);
Update $\mathbf{A}_2$ by SUnSAL algorithm;
Until Exceed the error tolerance or meet the maximum iteration number.
Combination: Merge $\mathbf{A}_1$ and $\mathbf{A}_2$ accordingly to obtain the abundance matrix $\mathbf{A}$.

IV. Experimental Results and Analysis

In this section, a series of experiments is designed to demonstrate the effectiveness of SRASU algorithm for HSIs. Here, we employ two simulated data sets and one real data set so as to evaluate its performance.

A. Evaluation Metrics

Three metrics are adopted to measure the accuracy of the extracted endmembers and their abundances that are widely used in [16] and [30]. They are spectral angle distance (SAD), root mean square error (RMSE), and signal-to-reconstruction error (SRE). The SAD measures the similarity between an estimated endmember $\bar{\mathbf{E}}_q$ and its corresponding ground truth $\mathbf{E}_q$, which is defined as follows:

$$\mathrm{SAD}\left(\mathbf{E}_q, \bar{\mathbf{E}}_q\right) = \arccos\left(\frac{\mathbf{E}_q^T \bar{\mathbf{E}}_q}{||\mathbf{E}_q|| ||\bar{\mathbf{E}}_q||}\right). \tag{15}$$

The smaller SAD value, the more similar $\bar{\mathbf{E}}_q$ and $\mathbf{E}_q$. The RMSE and SRE are used to evaluate the quality of estimated abundances, which are defined as follows:

$$\mathrm{RMSE}\left(\mathbf{A}, \bar{\mathbf{A}}\right) = \left(\frac{1}{P} ||\mathbf{A} - \bar{\mathbf{A}}||^2\right)^{\frac{1}{2}} \tag{16}$$

$$\mathrm{SRE}\left(\mathbf{A}, \bar{\mathbf{A}}\right) = 10 \log_{10}\left(\frac{E\left(||\mathbf{A}||^2\right)}{E\left(||\mathbf{A} - \bar{\mathbf{A}}||^2\right)}\right) \tag{17}$$

where $\mathbf{A}$ is the abundance of ground truth and $\bar{\mathbf{A}}$ is its estimated value. The smaller RMSE value corresponds to the better performance of unmixing. On the opposite, the higher SRE, the better the quality of abundance estimation.

B. Data Sets

The first simulated data set has $100 * 100$ pixels and 224 wavelengths, which was provided by Dr. M. D. Iordache and Prof. J. M. Bioucas-Dias. There are nine spectra randomly selected from the USGS spectral library, which are used to simulate the hyperspectral data set [54]. The abundances follow a Dirichlet distribution [55] whose images are shown in Fig. 2. Due to the good spatial homogeneity of this data set, it has been widely adopted for many unmixing algorithms [40], [56], [57]. In addition, Gaussian noise was added to the

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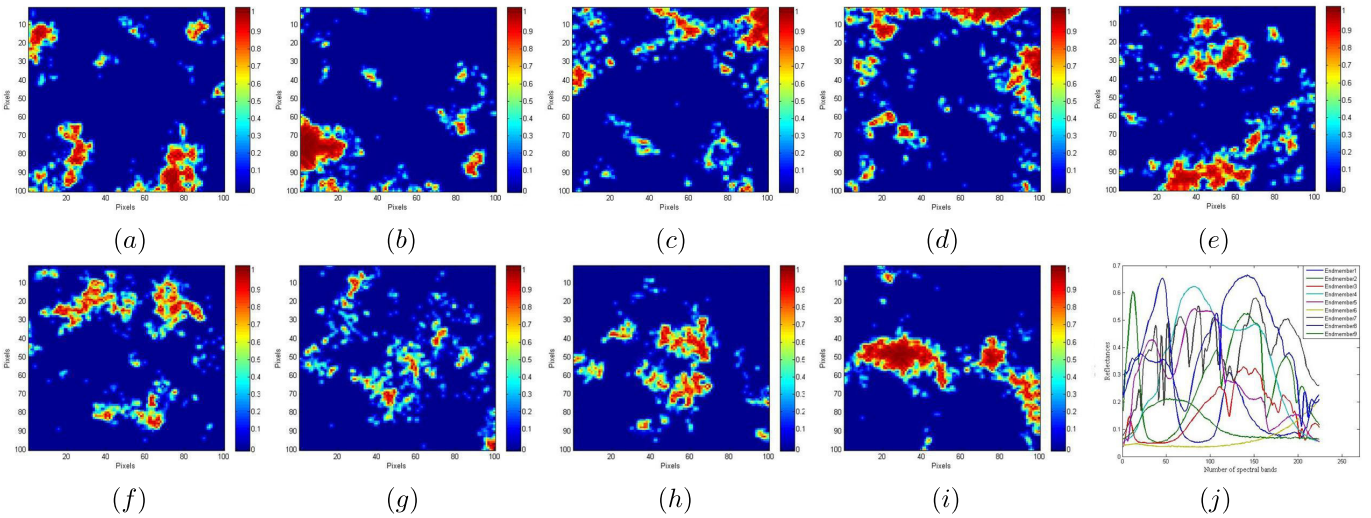


Fig. 2. True fractional abundances of simulated data set 1. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9. (j) Spectral signatures.

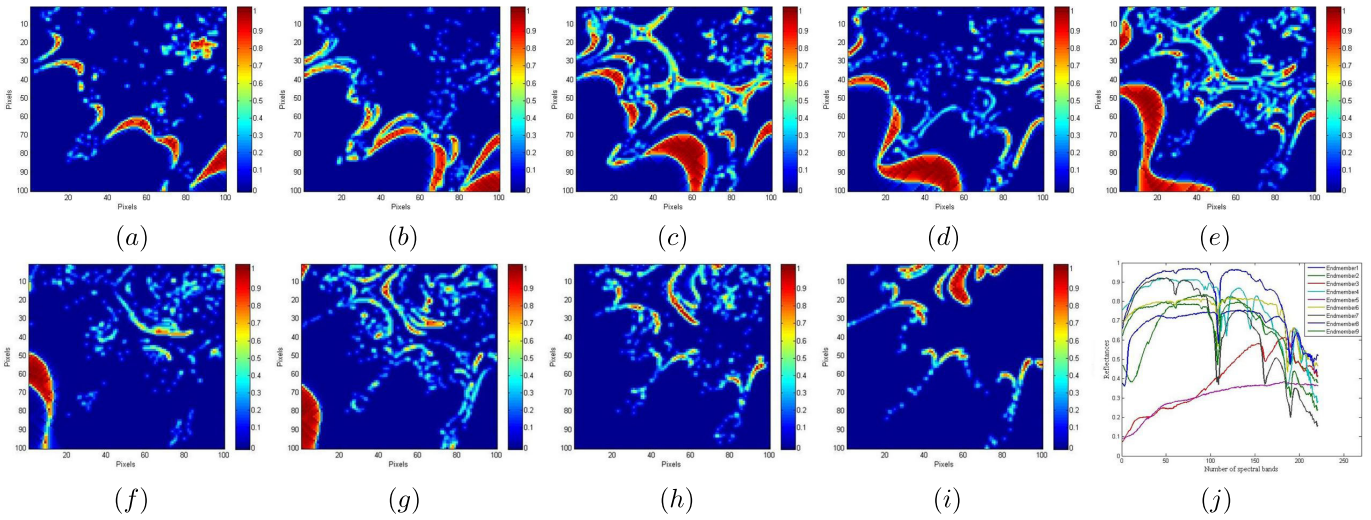


Fig. 3. True fractional abundances of simulated data set 2. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9. (j) Spectral signatures.

simulate data set with a given signal-to-noise ratio (SNR) that is defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{E[\mathbf{y}^T\mathbf{y}]}{E[\mathbf{n}^T\mathbf{n}]} \tag{18}$$

where $\mathbf{y}$ and $\mathbf{n}$ are the observation and noise of a pixel, respectively, and $E[]$ denotes the expectation operator. After obtaining the simulated data set, SNR $=$ 30 dB Gaussian noise was added.

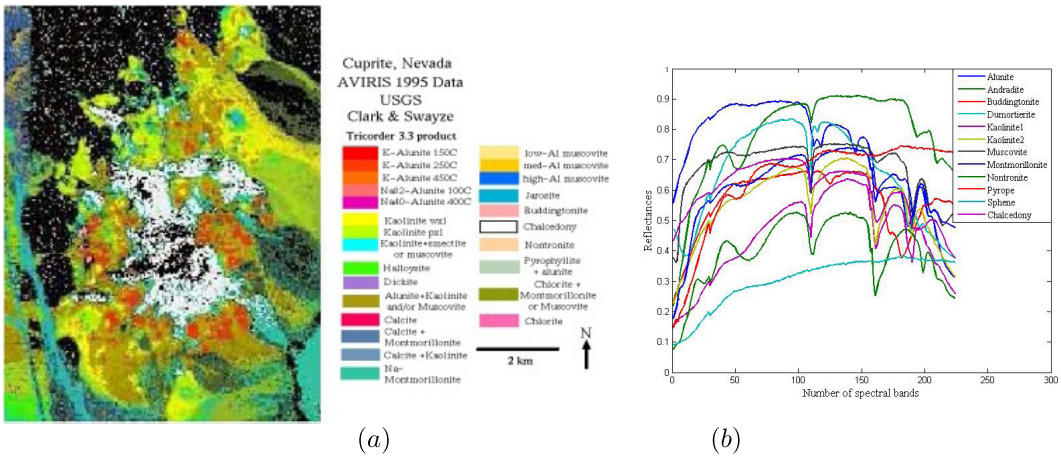
There are $100 * 100$ pixels and 221 wavelengths for the second simulated data set, which was considered as a standardized data set provided in the HyperMix tool [58] to test the spectral unmixing algorithms. Gaussian noise with an SNR of 30 dB was also added to this simulated data set. Fig. 3 shows the spectra selected from the spectral library and the true abundance maps that follow the LMMs together with the ASC and ANC.

The real data set is called Cuprite HSI captured by AVIRIS, which is a popular hyperspectral data used in the HU task [31], [32], [54]. The Cuprite data set has $250 * 191$ pixels, and each pixel has 224 channels ranging from 400 to 2500 nm. The spectral resolution and spatial resolution are separately up to 10 nm and 20 m, respectively. After removing channels 1 and 2, 105–115, 150–170, and 223 and 224 that have been effected by water vapor and atmospheric, it remains 188 channels. There is no available agreement for the number of endmembers in this data set. The frequently used and publicly recognized is 12 kinds of mineral types, namely, alunite, andradite, buddingtonite, dumortierite, kaolinite1, kaolinite2, montmorillonite, muscovite, nontronite, pyrope, sphene, and chalcedony. Fig. 4 shows the Cuprite data set and the corresponding reference spectral signatures chosen from the USGS library.

C. Compared Algorithms

In experiments, several related methods selected as the comparison algorithms are listed as follows.

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(a) (b)

Fig. 4. (a) Cuprite data set and (b) 12 different spectral curves of its endmembers.

SUnSAL method is one of the early sparse methods which is always selected as the comparison algorithm for unmixing.

$L_{1/2}$-NMF method is a typical unmixing method based on sparsity constraint.

GLNMF method is an approach based on manifold constraints, which could extract the graph information latent in hyperspectral data.

CoNMF method [32] is an NMF-based sparse unmixing algorithm, which adds spatial constraint for unmixing.

D. Experimental Settings

The first issue is the initialization method of matrices $\mathbf{E}$ and $\mathbf{A}$. In our experiment, VCA-FCLS, a basic method for endmember extraction and abundance estimation, is used as the initialization method for the two matrices. The VCA algorithm extracts the endmembers of whole HSIs as the initialization endmembers for these two regions, which considers global characteristics of endmembers signatures.

The stopping criterion is another important issue. We have two criteria for optimization, i.e., error tolerance and maximum iteration number. Once the error is within the limits of a predefined value set as the tolerance or the times of iteration meet the maximum iteration number, the operation ends. The maximum iteration number in our experiment is 500.

The next issue is about the $L_{1/2}$ sparsity operator, specifically the term $\mathbf{A}_1^{-1/2}$ of (10). To make the operation valid and improve the robustness of algorithm, there are two measures mentioned in [30]. If any entry in $\mathbf{A}_1$ is zero, a very small value will be added and those elements less than 10^{-4} will be updated by (13) without their corresponding term $(\lambda/2)\mathbf{A}_1^{-1/2}$.

The parameters setting is the last issue, and there are three sets of parameters need to be analyzed.

1) The parameter ε controls the influence of ASC over the abundance matrix. In practice, a more accurate result can be obtained by a large value of ε, which will make the convergence rate low. In order to achieve the desired tradeoff, a relatively small value is selected in the experiments of many papers [30], [31], [33]. As suggested in these papers, the parameter ε is set to be 15 here.
2) σ and μ are two parameters of weight function $\mathbf{W}$. The parameter σ and μ, respectively, control the width and height of function $\mathbf{W}$. Moreover, β is the balance parameter of manifold constraint. These three parameters are determined by experiments and analysis, which are defined next.
3) The parameters λ and α weight the contribution of sparse constraints on homogeneous regions and detail regions, respectively. Since the values of λ and α rely on the sparsity of material abundances which are unknown, similar to the method adopted in many articles [30], [31], [42], we use one possible estimator based on sparseness measure to calculate their values, which are given by

$$\lambda = \frac{1}{\sqrt{L}} \sum_{l} \frac{\sqrt{P_1} - ||\mathbf{y}_{1l}||_1/||\mathbf{y}_{1l}||_2}{\sqrt{P_1 - 1}} \tag{19}$$

$$\alpha = \frac{1}{\sqrt{L}} \sum_{l} \frac{\sqrt{P_2} - ||\mathbf{y}_{2l}||_1/||\mathbf{y}_{2l}||_2}{\sqrt{P_2 - 1}} \tag{20}$$

where $\mathbf{y}_{1l}$ and $\mathbf{y}_{2l}$, respectively, represent the given hyperspectral data of homogeneous regions and detail regions at band l.

E. Experiment on Simulated Data Set 1

First, we use the region generation method to adaptively obtain the sketch map, homogeneous region map, and detail region map shown in Fig. 5. From Fig. 5(a), it can be easily found that the sketch map consisting of sketch lines with directional characteristics can reflect the structure information of the original image. In addition, the areas where the sketch lines located indicate the areas with great spectral variance. These areas contain a lot of details, such as edges and textures, which are considered as the detail regions.

After getting the different regions of HSIs, unmixing is performed according to Algorithm 1. However, we need to determine the endmember number in advance, which is very

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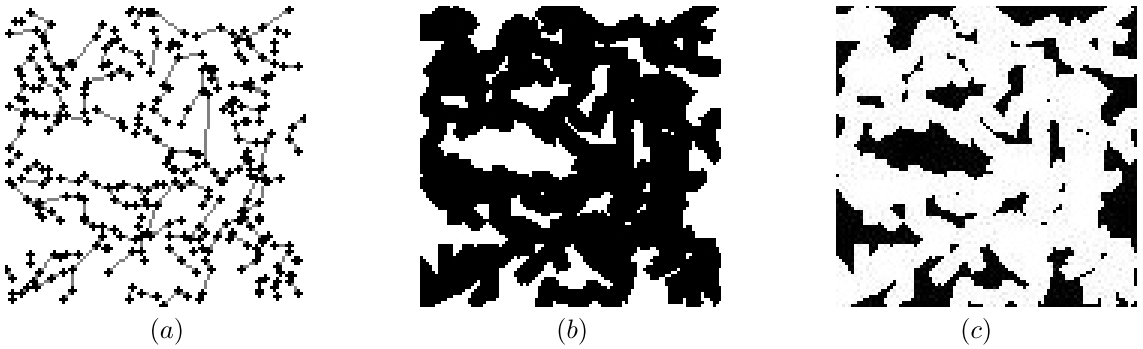


Fig. 5. Maps of sketch, homogeneous regions, and detail regions on simulated data set 1 with the white color. (a) Sketch map. (b) Homogeneous region map. (c) Detail region map.

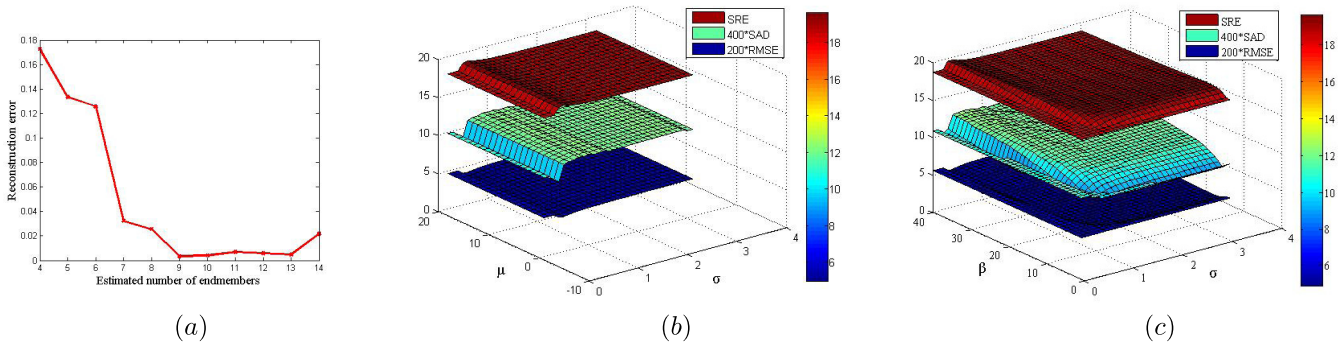


Fig. 6. Endmember number analysis by (a) reconstruction error, and the performances of SRASU with respect to (b) parameters σ and μ and (c) parameters σ and β in terms of SRE, SAD, and RMSE on simulated data set 1.

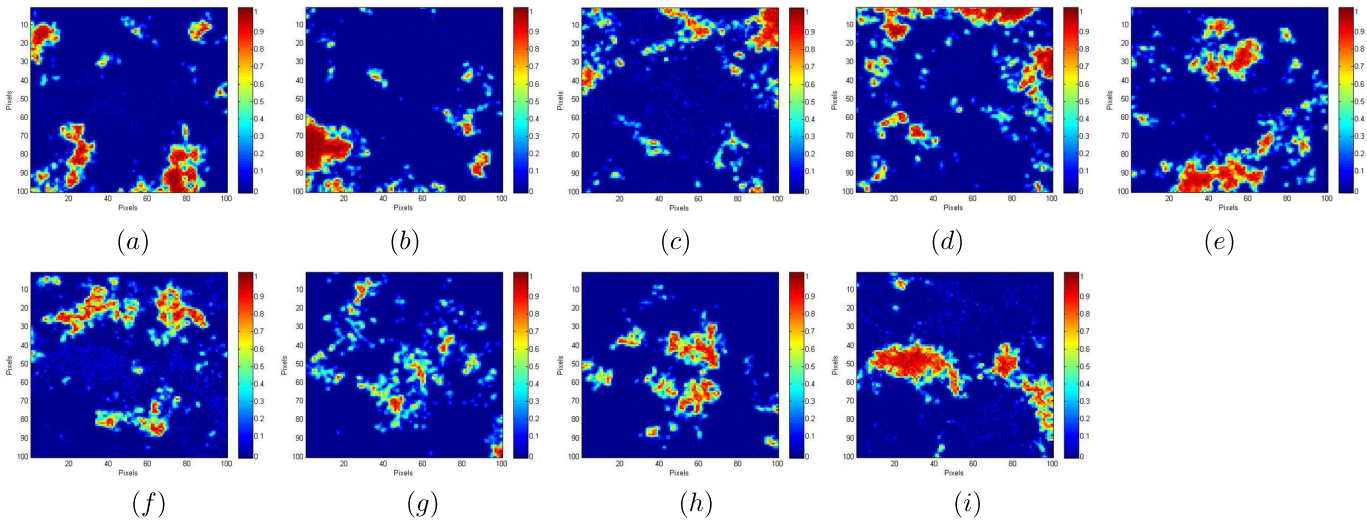


Fig. 7. Performances of SRASU with nine endmembers. (a) Endmember 1, (b) endmember 2, (c) endmember 3, (d) endmember 4, (e) endmember 5, (f) endmember 6, (g) endmember 7, (h) endmember 8, and (i) endmember 9 on simulated data set 1.

important to the task of HU. Let $||\mathbf{Y} - \bar{\mathbf{E}}\bar{\mathbf{A}}||^2$ be the reconstruction error for different numbers of endmembers [32]. Fig. 6(a) shows the relationship of endmember number and reconstruction error. It can be found that as the number of endmembers increases, the reconstruction error decreases gradually. When the endmember number is more than 8, the reconstruction error is small. Then, the error is stable with the endmember number increasing. Therefore, we choose 9 as the number of endmembers for simulated data set 1.

In addition, several parameters should be determined. In our model, λ and α are two parameters to control the sparse terms of homogeneous regions and detail regions whose values can be determined adaptively. We calculate their values by (19) and (20), which is one common measure adopted in many articles [30], [31], [42]. Moreover, the parameters β, σ, and μ are also analyzed. Fig. 6(b) and (c) shows the performance of SRASU with respect to these three parameters β, σ, and μ in terms of SRE, SAD, and RMSE. The SRE, SAD, and

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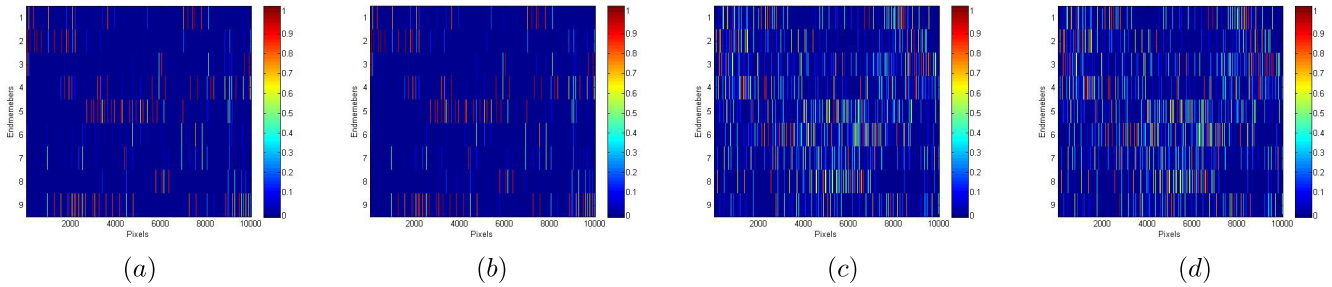


Fig. 8. Abundance matrices of homogeneous regions and detail regions on simulated data set 1. (a) Estimated abundance matrix of homogeneous regions. (b) Real abundance matrix of homogeneous regions. (c) Estimated abundance matrix of detail regions. (d) Real abundance matrix of detail regions.

RMSE adopted here are mean values for nine endmembers. In order to display the details of results clearly, we perform $400 * \text{SAD}$ and $200 * \text{RMSE}$. As shown in Fig. 6(b), with the increase in μ value, the values of SRE, SAD, and RMSE change slightly, which indicates that the proposed algorithm is insensitive to the parameter μ. Thus, in order to reduce the complexity of the algorithm, we set $\mu = 0$. Fig. 6(c) shows the variation of SRE, SAD, and RMSE with the parameters σ and β. Through observation, it can be found that when the value of σ is lower than 0.5, the SAD value is low. It means that the estimated endmembers are very close to the real values. The premise of the unmixing problem should be with complete and precise endmembers. Therefore, although the gap between the maximum and minimum values of SAD is small, σ is set to 0.3 to get the more precise endmembers and $\beta = 30$ corresponds to the better unmixing effect.

Fig. 7 plots the experimental results of SRASU. Compared with the abundance maps in Fig. 2, it can be easily seen that the unmixing results of our method are close to the real abundances, which illustrates the effectiveness of our method. Besides, sparsity degree comparisons of abundance values of two regions are also shown in Fig. 8. On one hand, by comparing the abundance values of homogeneous regions and detail regions, it can be easily seen that the sparsity of two regions is different. The abundance matrix of homogeneous regions is more sparse with fewer endmembers participated and the mixing in detail regions is more serious. On the other hand, the difference between the real abundance values and the estimated abundance values is small. To show the difference among the related unmixing methods, the error maps of comparisons and SRASU are shown in Fig. 9. They are separately SUnSAL, $L_{1/2}$-NMF, GLNMF, and CoNMF. Due to the limited space, the error maps of only four endmembers, i.e., endmembers 3, 6, 8, and 9, are shown in this article. In Fig. 9, the result of CoNMF is the worst, and the other four algorithm results are approximate. However, the overall color of the error maps obtained by our method is darker than the comparisons, indicating the small error in general, especially for Fig. 9(b) and (d). According to the abovementioned analysis, the proposed algorithm is efficient for HU.

Moreover, we record the quantitative analysis to show the difference among the comparisons and SRASU in Table I by three evaluation metrics, i.e., SRE, RMSE, and SAD. From

TABLE I

SRE, RMSE, AND SAD VALUES OF SRASU AND COMPARISONS FOR SIMULATED DATA SET 1

		SUnSAL	$L_{1/2}$-NMF	GLNMF	CoNMF	SRASU
SRE	Average	18.3438	18.4285	18.8249	8.1171	**19.2661**
	EM1	22.9949	23.1764	23.2342	9.1989	**24.2065**
	EM2	21.7783	21.8002	21.8041	9.1098	**22.7935**
	EM3	15.6364	15.6927	15.6948	6.500	**16.1633**
	EM4	22.4679	22.6526	21.8336	10.0991	**23.4339**
	EM5	22.4465	22.4335	22.1606	9.8897	**23.9045**
	EM6	15.7538	15.8401	16.0011	7.0671	**16.2969**
	EM7	22.3009	**22.3507**	20.3241	9.5711	22.0890
	EM8	20.2544	20.3554	**20.9069**	8.3229	*20.6776*
	EM9	14.4270	14.5232	**16.0224**	5.8633	*15.8403*
RMSE	Average	0.0280	0.0278	0.0273	0.0973	**0.0254**
	EM1	0.0184	0.0180	0.0179	0.0899	**0.0160**
	EM2	0.0196	0.0196	0.0196	0.0843	**0.0174**
	EM3	0.0394	0.0392	0.0392	0.1129	**0.0371**
	EM4	0.0218	0.0213	0.0234	0.0904	**0.0195**
	EM5	0.0220	0.0220	0.0227	0.0934	**0.0186**
	EM6	0.0394	0.0390	0.0383	0.1070	**0.0370**
	EM7	0.0147	**0.0146**	0.0184	0.0635	0.0150
	EM8	0.0221	0.0219	**0.0205**	0.0874	*0.0211*
	EM9	0.0550	0.0543	**0.0457**	0.1473	*0.0467*
SAD	Average	0.0315	0.0244	0.0276	0.0315	**0.0235**
	EM1	0.0231	**0.0154**	0.0302	0.9324	*0.0196*
	EM2	0.0193	0.0113	0.0151	0.8367	**0.0100**
	EM3	0.0475	0.0426	0.0380	0.0475	**0.0359**
	EM4	0.0263	0.0113	0.0108	0.4074	**0.0084**
	EM5	0.0228	0.0189	0.0168	0.5091	**0.0156**
	EM6	0.0714	**0.0588**	0.0811	0.9601	*0.0591*
	EM7	0.0110	0.0102	**0.0073**	0.3990	0.0144
	EM8	0.0089	**0.0070**	0.0077	0.7562	0.0100
	EM9	0.0533	0.0438	0.0414	0.8657	**0.0385**

Table I, we find that the proposed method can get the highest SRE and the lowest RMSE values for most endmembers, indicating its effectiveness for abundance estimation. Furthermore, the differences between the obtained endmembers by the proposed method and real endmembers are small. In general, since we take into consideration the different sparsity degrees of two regions and the intrinsic manifold structure, the SRASU algorithm can obtain good results for unmixing.

F. Experiment on Simulated Data Set 2

We first present in Fig. 10 the maps of sketch and two different regions on simulated data set 2 by a region generation method. The sketch map consisting of sketch lines contains a lot of details and there are fewer homogeneous areas in this data set. Due to the vision information, the segmentation

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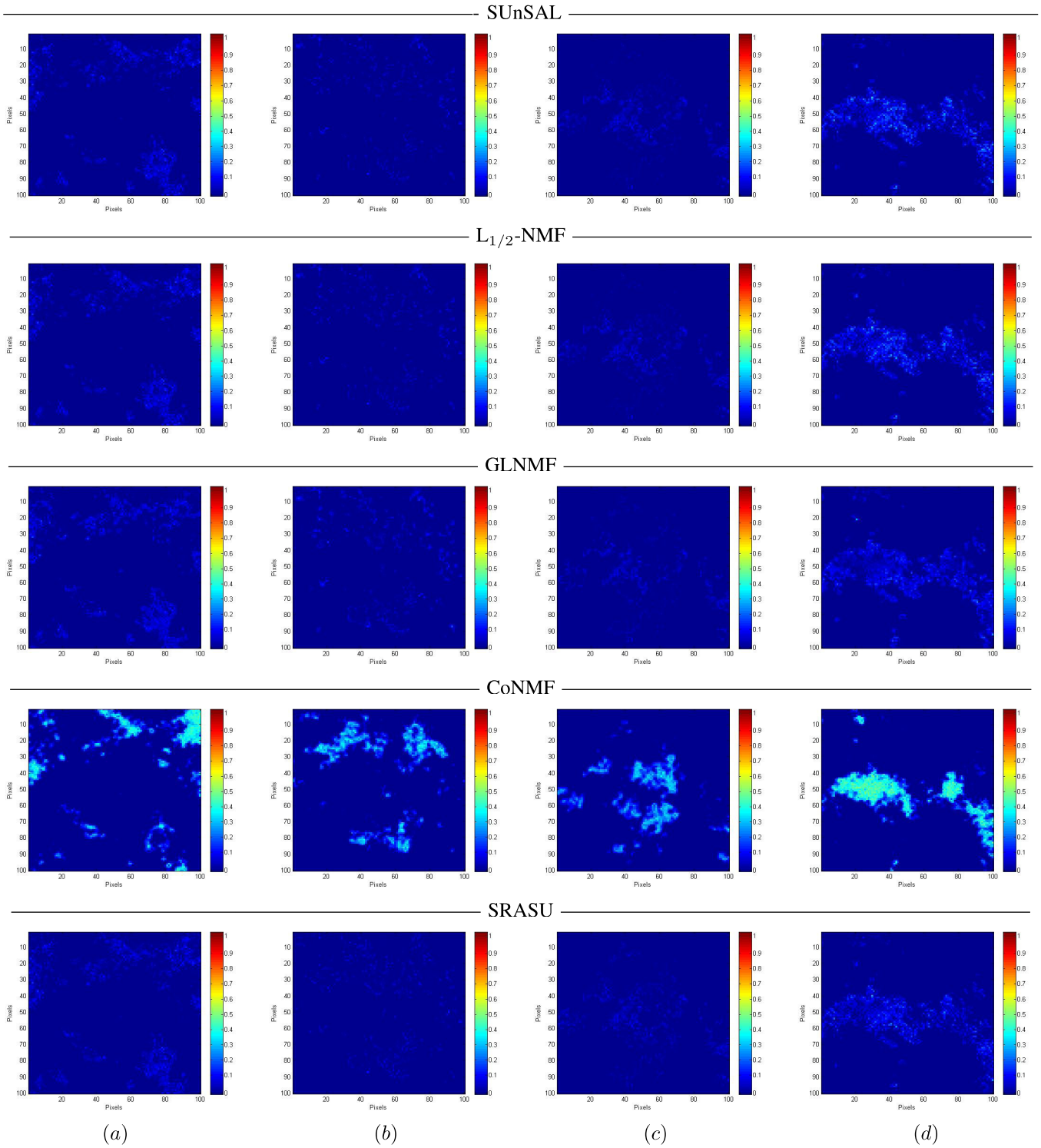


Fig. 9. Error maps of abundances obtained by the comparisons and SRASU with four endmembers. (a) Endmember 3, (b) endmember 6, (c) endmember 8, and (d) endmember 9 on simulated data set 1.

lays the foundation for the later unmixing. Then, by the rectangular window with the direction, the image can be divided into homogeneous regions and detail regions shown in Fig. 10(b) and (c).

Similarly, for simulated data set 2, it is necessary to estimate the endmember number in advance which is exhibited in Fig. 11(a). As the number of endmembers increases, the reconstruction error decreases in general. When the number of endmembers is 9, the reconstruction error is very small and tends to be stable. Therefore, 9 is selected as the number of endmembers. We also analyze the parameters of the proposed method for simulated data set 2. The parameters λ and α are

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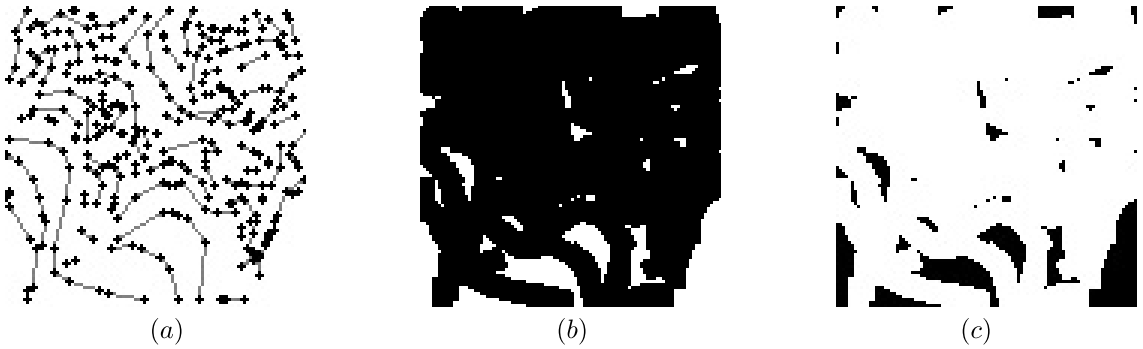


Fig. 10. Maps of sketch, homogeneous regions, and detail regions on simulated data set 2 with the white color. (a) Sketch map. (b) Homogeneous region map. (c) Detail region map.

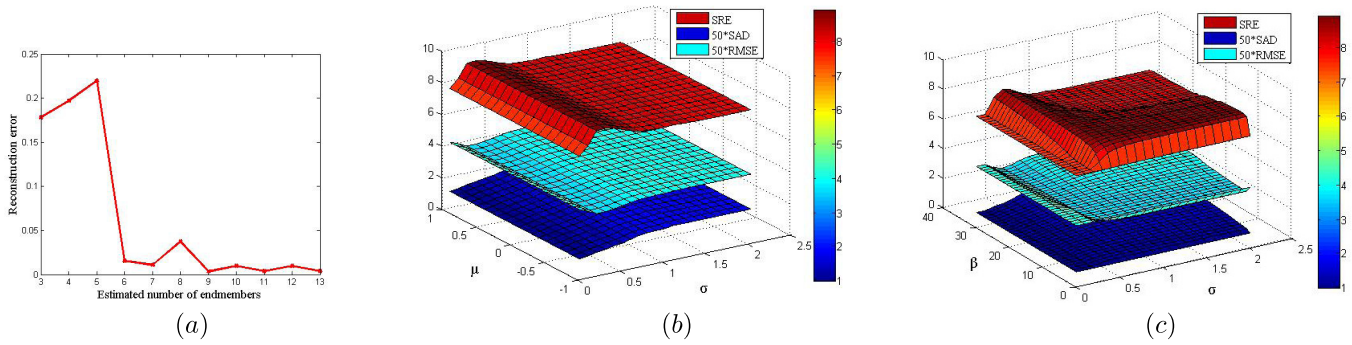


Fig. 11. Endmember number analysis by (a) reconstruction error, and the performances of SRASU with respect to (b) parameters σ and μ and (c) parameters σ and β in terms of SRE, SAD, and RMSE on simulated data set 2.

calculated by (19) and (20). Besides, the curves of other three parameters of our method, i.e., β, σ, and μ are shown in Fig. 11(b) and (c). The evaluation metrics adopted here are the average SRE, SAD, and RMSE calculated by (15)–(17). Fig. 11(b) and (c), respectively, shows the effects of parameters σ–μ and σ–β, on SRE, SAD, and RMSE. Similarly, the parameter μ is set to 0 due to its stability. On one hand, as is known to all, the precise endmembers are necessary for unmixing problem. The proposed algorithm obtains good capability of endmember extraction when the value of σ is less than 0.5, as shown in Fig. 11(b) and (c). On the other hand, the three curves of SRE, SAD, and RMSE tend to stable when the value of β is more than 10. To achieve a desired result, the parameter β is set to be 20 in the experiment on simulated data set 2. The proposed algorithm with low SAD value, low RMSE value, and high SRE value is achieved with $\sigma = 0.2$.

After determining the model parameters, we conduct experiments on simulated data set 2. Moreover, to further evaluate the effectiveness of the proposed algorithm, we compare our method against SUnSAL, $L_{1/2}$-NMF, GLNMF, and CoNMF, whose abundance maps are drawn in Fig. 12. Meanwhile, due to the limited space, article only depicts the abundance maps of four endmembers, including endmembers 2, 4, 7, and 9. By comparing the abundance maps in Fig. 12, it can be seen that the unmixing results are similar except for CoNMF. Due to the complexity of data set scene, it is relatively difficult for unmixing. The main difference of the comparisons and SRASU reflects in some details. For example, in the area around [80, 80] and the upper right corner of abundance maps for endmember 2, the color of abundance map by SRASU is lighter, which is closer to the real abundance map. Compared with simulated data set 1, the scene of this data set is complicated with fewer homogeneous regions. The sparse difference of abundances of two regions in the image is small, which makes it difficult to reflect the advantages of the proposed algorithm. It is actually a limitation of SRASU whose performance will be affected by scene complexity.

Moreover, for quantitative analysis, the unmixing results of the proposed method and the comparisons are listed in Table II, including the metrics of SRE, RMSE, and SAD. The SRASU algorithm achieves most of the best and second best results in terms of SRE, SAD, and RMSE. Since we utilize the different sparsity features of two regions and exploit the latent structure of data, the proposed method takes advantage of the spatial–spectral information of HSIs. In general, our method achieves better results than comparisons.

G. Experiment on Cuprite Data Set

We now turn our attention to the real-world data set, which is given by the well-known AVIRIS Cuprite data set in west-central Nevada in Fig. 4. After using a region generation algorithm, the sketch map and two different region maps of the Cuprite data set are shown in Fig. 13. From this figure, we note that due to the complexity of this data set, the sketch map extracts many edges and even some tiny details by considering the vision information. The homogeneous regions and detail

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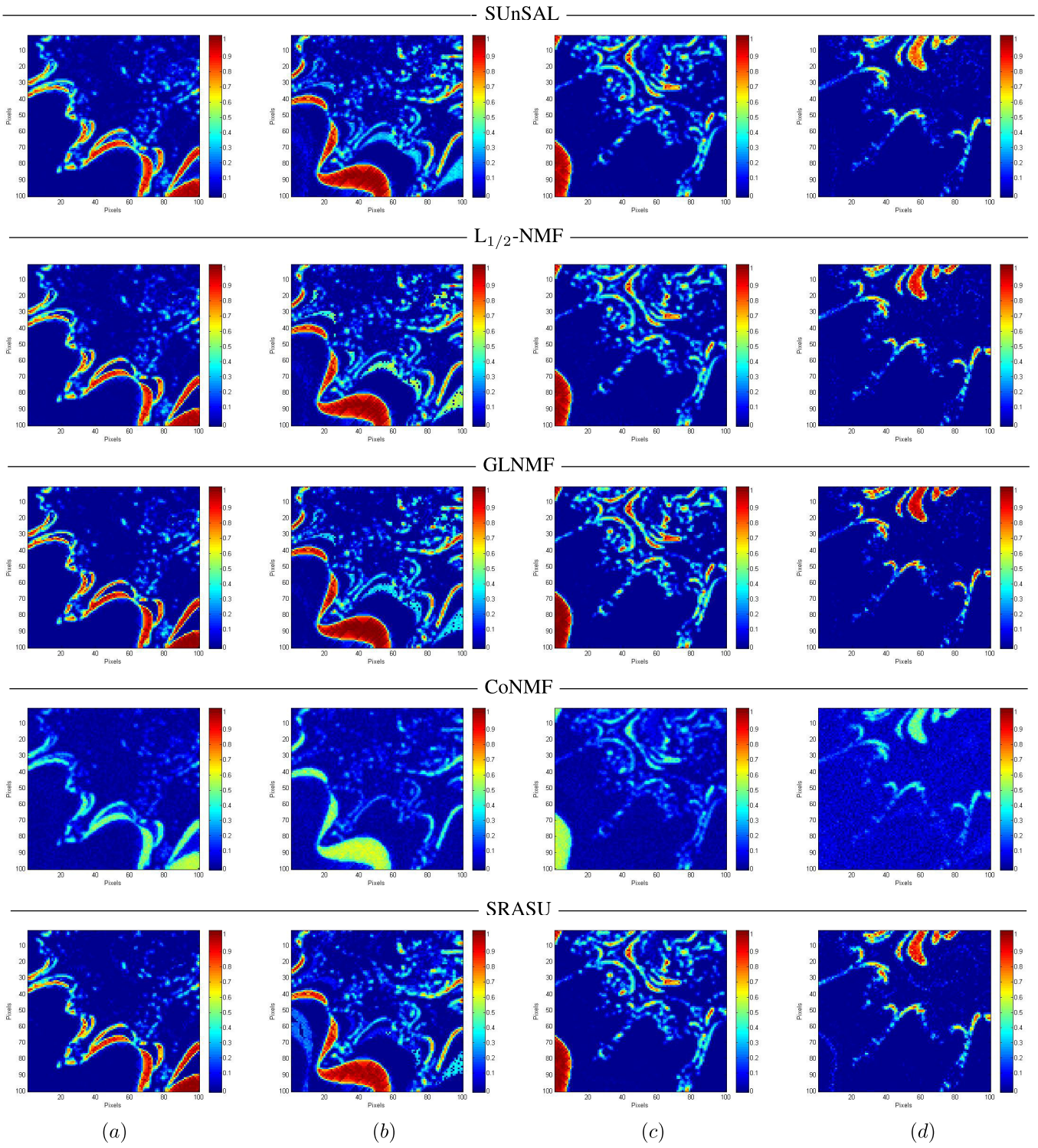


Fig. 12. Performances of the comparisons and SRASU with four endmembers. (a) Endmember 2, (b) endmember 4, (c) endmember 7, and (d) endmember 9 on simulated data set 2.

regions of the Cuprite data set are segmented well, which is important for the proposed unmixing method.

As for the simulated data set, we use the same method to estimate the number of endmembers for the real data set, as shown in Fig. 14. It can be observed that when the number of endmembers is 12–14, the reconstruction error is small and the smallest value is obtained with 12 endmembers. According to some existing analyses [16], [31], we know that there are only tiny differences between some of the spectra of the same mineral with different chemical compositions, which is related to the spectral variability and the presence of mineral alterations. Therefore, the number of endmembers

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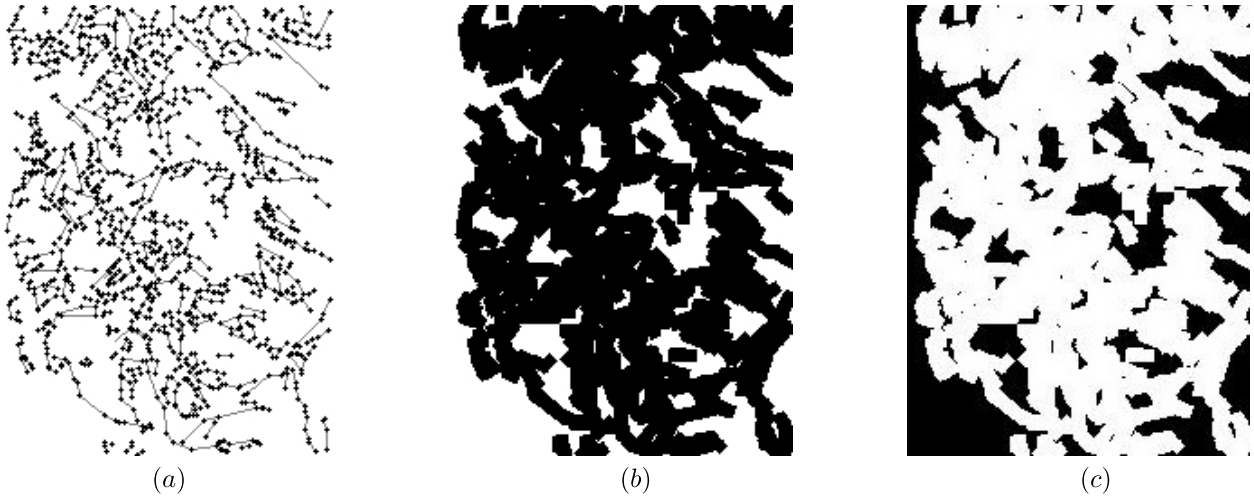


Fig. 13. Maps of sketch, homogeneous regions, and detail regions on the Cuprite data set with the white color. (a) Sketch map. (b) Homogeneous region map. (c) Detail region map.

TABLE II
SRE, RMSE, AND SAD VALUES OF SRASU AND COMPARISONS FOR SIMULATED DATA SET 2

		SUnSAL	$L_{1/2}$-NMF	GLNMF	CoNMF	SRASU
SRE	Average	6.5341	6.5516	7.4367	6.4411	**8.0181**
	EM1	1.0014	0.5283	1.0648	3.1874	**4.5018**
	EM2	**17.0130**	15.2560	14.3288	7.7389	*16.7888*
	EM3	17.1249	17.5039	16.8807	7.5717	**18.4751**
	EM4	11.4791	9.0942	11.4509	7.9929	**12.2454**
	EM5	6.2056	7.3655	**10.1539**	6.9950	*8.8782*
	EM6	-1.2513	-1.0526	-0.3615	**5.4740**	-0.8912
	EM7	17.1086	18.1205	15.6465	7.6430	**18.6905**
	EM8	10.5894	**10.7163**	10.0603	4.6387	10.2120
	EM9	11.4303	**13.3676**	12.9825	4.3740	12.8040
RMSE	Average	0.0862	0.0861	0.0828	0.1152	**0.0775**
	EM1	0.1788	0.1888	0.1775	0.1390	**0.1195**
	EM2	**0.0345**	0.0422	0.0469	0.1003	*0.0354*
	EM3	0.0451	0.0432	0.0464	0.1355	**0.0386**
	EM4	0.0766	0.1099	0.0769	0.1145	**0.0702**
	EM5	0.1534	0.1342	**0.0974**	0.1401	*0.1128*
	EM6	0.2214	0.2164	0.1998	**0.1021**	0.2124
	EM7	0.0324	0.0288	0.0383	0.0962	**0.0270**
	EM8	0.0492	**0.0484**	0.0522	0.0975	0.0513
	EM9	0.0495	**0.0396**	0.0414	0.1116	0.0423
SAD	Average	0.0213	**0.0200**	0.0235	0.1240	*0.0212*
	EM1	0.0663	0.0528	0.0415	0.0413	**0.0304**
	EM2	0.0148	0.0073	0.0105	0.0863	**0.0069**
	EM3	0.0162	**0.0124**	0.0273	0.7171	0.0185
	EM4	0.0131	0.0119	0.0115	0.0690	**0.0066**
	EM5	**0.0183**	0.0195	0.0292	1.5181	0.0275
	EM6	0.2957	0.2959	0.2942	**0.0453**	*0.2644*
	EM7	0.0085	0.0069	0.0206	0.1632	**0.0067**
	EM8	0.0312	0.0279	0.0244	0.0762	**0.0234**
	EM9	0.0156	**0.0107**	0.0145	0.0722	0.0146

for the Cuprite data set is set to be 12 as the setting in most articles. Since there is no ground truth for Cuprite data set, the parameters are determined based on experiments and the previous analyses. λ and α are calculated by (19) and (20). In addition, parameter μ for this data set is set to be 0 due to its small effect.

The obtained unmixing results are shown in Fig. 15. It can be seen that the different ground types are well distinguished in general according to Fig. 4. Moreover, the abundance matrices for homogeneous regions and detail regions are exhibited in Fig. 16. From it, we can easily observe that the sparsity

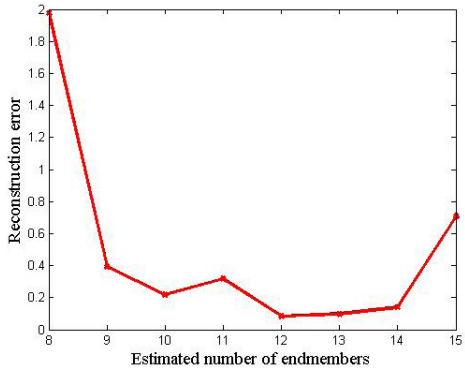


Fig. 14. Endmember number analysis by reconstruction error on the Cuprite data set.

TABLE III
SAD VALUES OF SRASU AND COMPARISONS FOR THE CUPRITE DATA SET

	SUnSAL	$L_{1/2}$-NMF	GLNMF	CoNMF	SRASU
Average	0.0833	0.0808	0.0804	0.1431	**0.0799**
Alunite	**0.0971**	0.1045	0.1087	0.4741	0.1191
Andradite	0.1701	0.1718	0.1634	**0.1346**	*0.1433*
Buddingtonite	0.1215	0.1128	**0.1117**	0.2084	0.1138
Dumortierite	0.2030	0.2005	0.1984	0.4117	**0.1891**
Kaolinite1	0.0896	0.0852	0.0842	0.2506	**0.0787**
Kaolinite2	**0.0914**	0.1073	0.1160	0.3485	0.1511
Muscovite	0.0942	0.0862	**0.0847**	0.1164	*0.0850*
Montmorillonite	0.0625	0.0632	0.0630	0.1155	**0.0616**
Nontronite	**0.1052**	0.1101	0.1103	0.3488	0.1204
Pyrope	0.1207	0.0835	0.0719	1.4144	**0.0719**
Sphene	0.1758	0.1740	0.1725	1.1747	**0.1687**
Chalcedony	**0.1789**	0.1799	0.1839	0.4468	0.1953

degrees of two kinds of regions are obviously different. The abundances matrix of homogeneous regions is much sparser than that of the detail regions, and there are more endmembers participated in detail regions.

Furthermore, we also compare the results of the proposed algorithm with other related algorithms. The comparison abundance maps are shown in Fig. 17. Due to the limited space, only three common endmembers are displayed. From Fig. 17, we find that the results of SRASU are close to Fig. 4. For

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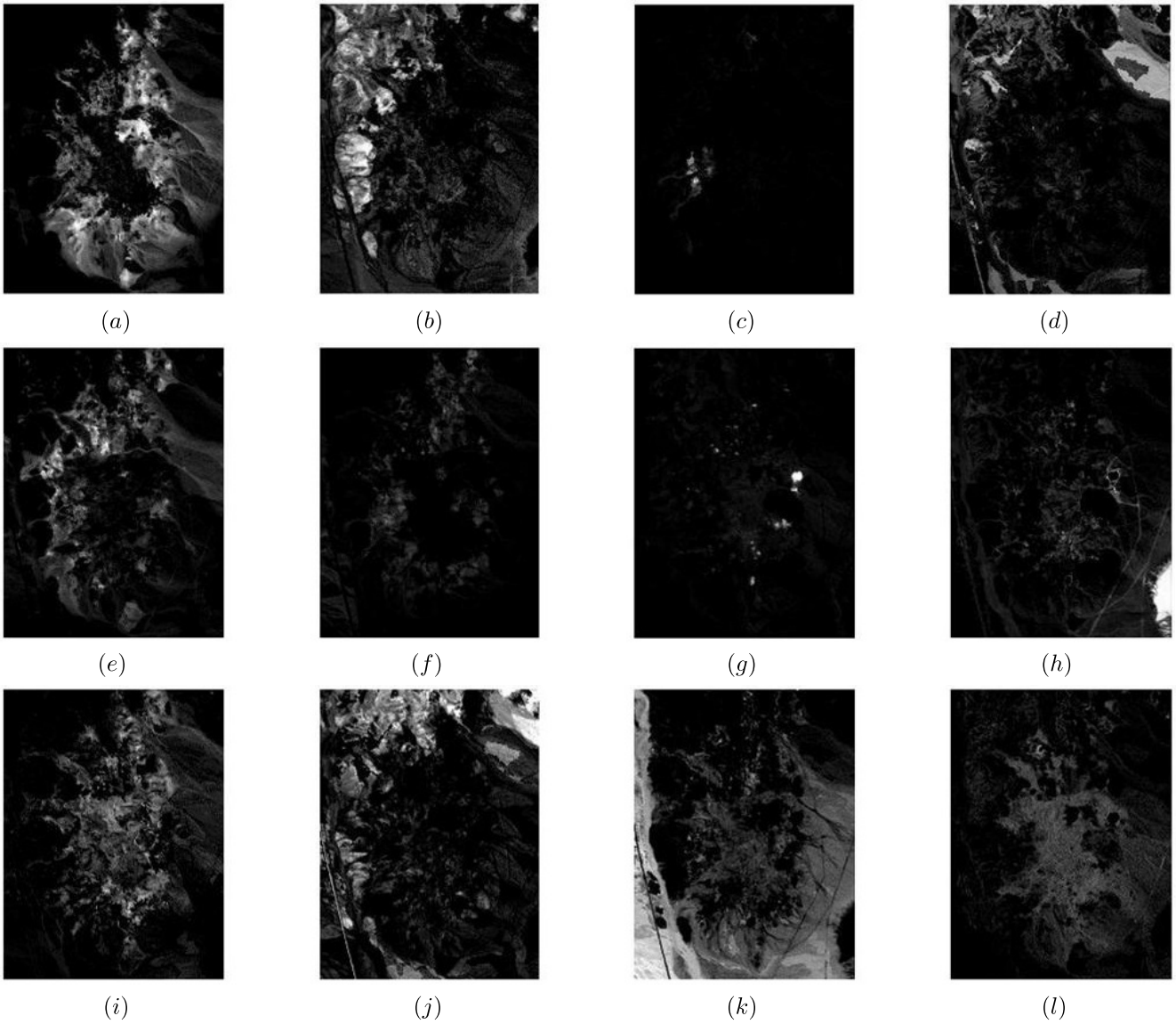


Fig. 15. Performances of SRASU with 12 endmembers. (a) Alunite, (b) andradite, (c) buddingtonite, (d) dumortierite, (e) kaolinite1, (f) kaolinite2, (g) muscovite, (h) montmorillonite, (i) nontronite, (j) pyrope, (k) sphene, and (l) chalcedony on the Cuprite data set.

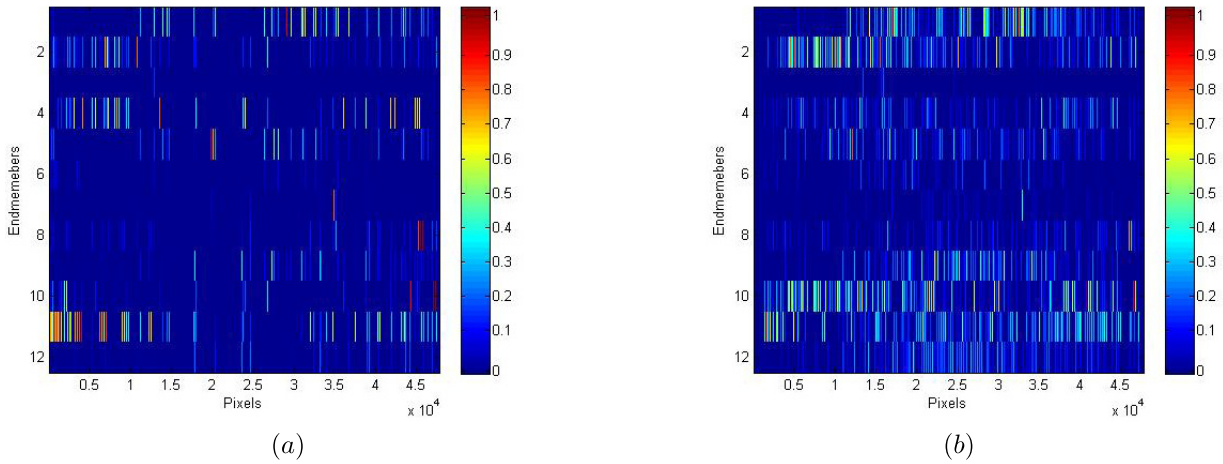


Fig. 16. Estimated abundance matrices of (a) homogeneous regions and (b) detail regions on the Cuprite data set.

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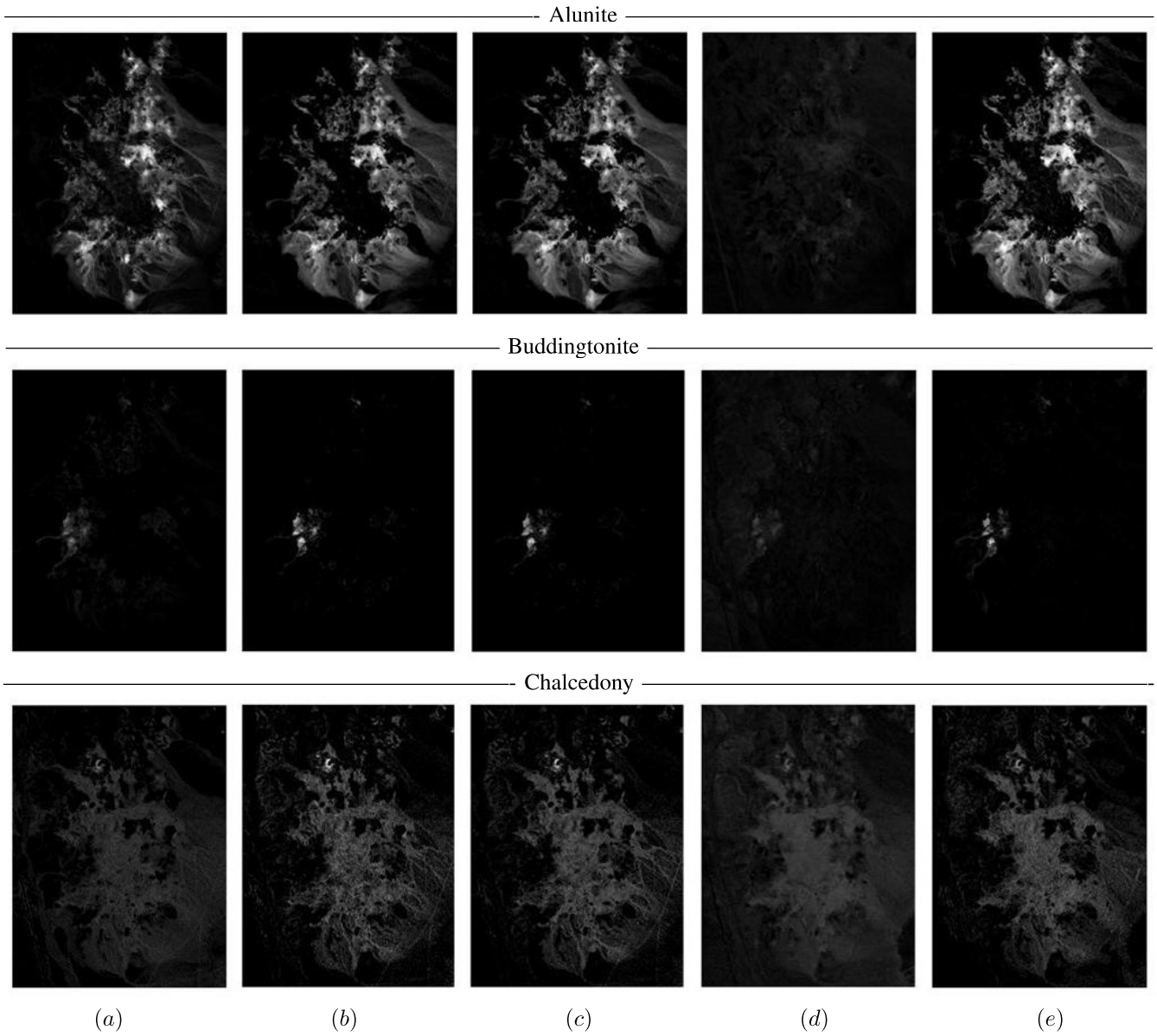


Fig. 17. Performances of comparisons. (a) SUnSAL, (b) $L_{1/2}$-NMF, (c) GLNMF, (d) CoNMF, and the proposed method (e) SRASU with three endmembers, including alunite, buddingtonite, and chalcedony on the Cuprite data set.

example, in terms of results for buddingtonite, the abundance maps obtained by SRASU are much clearer in the background than the comparisons. Meanwhile, Table III records the SAD values of the endmembers obtained by different methods. The proposed algorithm gets most of the minimum results, proving the good performance of the proposed method. Based on the abovementioned analyses, it can be concluded that the SRASU algorithm is effective for unmixing.

V. Conclusion

In this article, inspired by the sparsity difference in spatial of HSIs, a novel method called SRASU applied to HSI has been proposed. The assumption of this article is that most pixels in the image are mixed, and the pixels in detail regions are more likely to be heavily mixed with more endmembers participated than in homogeneous regions. Therefore, the sparsity degrees of the abundances for these two regions are obviously different, and the sparsity degree of abundance in homogeneous regions is relatively high. Thus, we first adaptively differentiate the homogeneous regions and detail regions by a region generation algorithm based on the sketch map, which makes full use of the vision information in the HSIs. Then, we separately constrain the abundances of these two regions by different sparse regularizers with a proposed manifold regularizer. The encouraged results illustrate the effectiveness of SRASU. In general, our method utilizes the spatial–spectral information of HSIs and exploits the latent structure of data. However, our algorithm also has a limitation. It only studies the situation that most pixels in the HSI are mixed and the mixed pixels and pure pixels exist simultaneously. It might be affected by the scene complexity of the data set and when the scene of the data set is complicated with fewer homogeneous regions, the performance of our method is similar to the available methods without splitting. In the future, more works will be devoted to the unmixing studies for complex scenarios and the cases with highly mixed data.

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